Quiz 4 (*Nov. 5, Marks: 10, Duration: 20 mins*) **COL703/7203, Jul-Nov 2026**

Name: **Entry No.:**

1. **[1 marks]** Write down a formula in first order logic with equality that captures the requirement that a unary predicate P holds for exactly three different elements.

   Ans:

   $\neg(a = b) \wedge \neg(b = c) \wedge \neg(a = c) \wedge P(a) \wedge P(b) \wedge P(c) \wedge \forall x\ (P(x) \rightarrow ((x = a) \vee (x = b) \vee (x = c)))$

2. Consider the following structures over a signature with a single binary relation symbol R. Give a predicate logic sentence that is satisfied by $\mathcal{A}$ but not by $\mathcal{B}$.

   $$U_{\mathcal{A}} = \mathbb{Q} \text{ and } R_{\mathcal{A}} = \{(n, m) \in \mathbb{Q} \times \mathbb{Q} \quad \text{such that} \quad n < m\}$$
   $$U_{\mathcal{B}} = \mathbb{Z} \text{ and } R_{\mathcal{B}} = \{(n, m) \in \mathbb{Z} \times \mathbb{Z} \quad \text{such that} \quad n < m\}$$

   Ans: $\forall x \forall y\ (R(x, y) \rightarrow \exists z\ (R(x, z) \wedge R(z, y)))$

3. **[1 marks]** Consider the following predicate-logic sentences.

   ϕ_1: $\forall x\ P(x, x)$
   ϕ_2: $\forall x \forall y\ (P(x, y) \rightarrow P(y, x))$
   ϕ_3: $\forall x \forall y \forall z\ (P(x, y) \wedge P(y, z) \rightarrow P(x, z))$

   These sentences express that P is reflexive, symmetric, and transitive.

   Show that transitivity is not semantically entailed by the other two properties. In other words, give a model (an assignment) that satisfies ϕ_1 and ϕ_2, but does not satisfy ϕ_3.

   Ans:

   Let the universe be the set $\{a, b, c\}$. And let the interpretation for P be as follows:

   $$P := \{(a, a), (b, b), (c, c), (a, c), (a, b), (b, a), (c, a)\}$$

   Note that P is reflexive and symmetric, but not transitive – because (b, a) and (a, c) are in P, but (b, c) is not.